

PAULA MAGDY

Software Engineer

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PROFILE

Full-Stack Software Engineer with production experience across TypeScript (Next.js/NestJS), PHP (Laravel), and Python (Django/Flask) on AWS, Azure, and Huawei Cloud. AWS Certified Solutions Architect contributing across data modeling, API design, CI/CD, monitoring, and security, with a track record of measurable gains in API performance and deployment speed.

PROFESSIONAL EXPERIENCE

Challenge Group — Full-Stack Software Engineer

Jun 2025 – Present

Cairo, Egypt • AWS/Huawei Cloud partner delivering production platforms for government and enterprise clients across Egypt, KSA and the UAE.

Contributed to multiple production platforms as part of cross-functional Agile delivery teams:

- Developed backend services for government and enterprise platforms serving 5,000+ concurrent users — domain-driven NestJS modules and Laravel APIs over PostgreSQL/Prisma and MySQL — applying targeted indexing and query tuning that cut API latency by ~35% under production load.
- Moved deployments to container-first CI/CD — Docker → ECR → ECS via CodeBuild/CodePipeline — cutting deployment time from ~20 minutes to under 5.
- Built the request-and-bid marketplace flow with escalation handling and an in-app wallet on Paymob, letting users pay and withdraw funds without leaving the platform.
- Delivered real-time and messaging features — WebSocket chat (Laravel Reverb), voice/SMS via Twilio, FCM push notifications, and S3-backed media storage — giving users live chat, calls, and alerts in Arabic and English.
- Hardened API layers across multiple products with Sanctum/JWT authentication, RBAC, attribute-based routing, and self-validating DTOs, rejecting unauthorized and malformed requests before they reached business logic.
- Provisioned and maintained AWS infrastructure (VPC, EC2, RDS, S3, CloudFront, IAM, Secrets Manager) for 5+ production environments, and independently designed and ran the Azure API Management layer.
- Instrumented services with CloudWatch monitoring and structured logging, and maintained automated test coverage (Pest, PHPUnit, Jest, Vitest) that runs in CI to catch regressions before release.
- Built Next.js/React interfaces — onboarding and results-entry flows, role-based admin panels (Filament/Livewire), and live-updating leaderboards — on a shared, responsive component set reused across products.
- Independently designed, built, and shipped a serverless Amazon Lex chatbot on AWS Lambda, integrated with Twilio, automating customer-support conversations and cutting the manual triage load on the support team.

Stack: NestJS, Laravel, Next.js, React, TypeScript, PHP, Python, PostgreSQL, Prisma, MySQL, Redis, Filament (Livewire), Sanctum, Reverb (WebSockets), Twilio, Paymob, Firebase Cloud Messaging (FCM), Docker, AWS (ECS, ECR, Lambda, Lex, RDS, S3, CloudFront, CloudWatch, Secrets Manager), Microsoft Azure, Azure API Management, CI/CD, Pest, PHPUnit.

Information Technology Institute (ITI) — Full-Stack Engineer

Oct 2024 – Jun 2025

Cairo, Egypt • Government-sponsored intensive software-engineering program (~1,200 hours)

- Built full-stack applications across two backend ecosystems — Node.js/Express and Python (Django, Flask) — with React front-ends, secure REST APIs, structured authentication flows, and role-based access control.
- Modeled both relational (SQL) and NoSQL databases, applying normalization and indexing to keep queries efficient as datasets grew.
- Containerized services with Docker and deployed to AWS, applying system-design fundamentals and working in Agile sprints from design through delivery.

Freelance — Front-End Developer

Oct 2023 – Oct 2024

Cairo, Egypt

- Built and shipped a production-ready e-commerce storefront in React — a responsive, cross-device UI wired to REST APIs for the product catalog and cart.
- Architected a reusable component library with centralized state via React's Context API, removing prop-drilling and keeping catalog and cart state consistent as the product range and traffic grew.
- Improved rendering performance and cut page-load times through render optimization and efficient asset loading, keeping large product listings smooth as the catalog scaled.

SKILLS

Languages: TypeScript, JavaScript, Python, PHP, SQL, C++

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, CSS, HTML, Livewire, Filament, Blade

Backend: Node.js, NestJS, Express.js, Laravel, Django, Flask, REST APIs, WebSockets, Reverb, Serverless, JWT, RBAC

Databases & ORMs: PostgreSQL, MySQL, MongoDB, Redis, Prisma ORM, Eloquent ORM, Query Optimization

Cloud & DevOps: AWS, Huawei Cloud, Microsoft Azure, Docker, Kubernetes, CI/CD

Testing & Practices: Vitest, Jest, PHPUnit, Pest, React Testing Library, Playwright, Domain-Driven Design, Agile/Scrum

CERTIFICATIONS

Huawei HCCDP – Solution Architectures

Jun 2026

Professional-tier certification in end-to-end solution design on Huawei Cloud — architecting for high availability, security, and cost efficiency across the platform's compute (ECS), storage (OBS), networking (VPC/ELB), and managed-database services.

AWS Certified Solutions Architect – Associate (SAA-C03)

Jun 2026

Validates designing fault-tolerant, scalable, and secure AWS architectures — VPC networking, EC2/Lambda compute, S3 and RDS/DynamoDB data layers, IAM, and multi-AZ high availability and disaster recovery.

Huawei HCCDA – Tech Essentials

May 2026

Associate-level foundation in Huawei Cloud — core compute, storage, and networking services, application deployment, and the platform's developer toolchain.

AWS Certified Cloud Practitioner

Apr 2026

Foundational AWS certification spanning cloud concepts and economics, the shared-responsibility security model, pricing and billing, and the platform's core service categories.

EDUCATION

B.Sc., Computer Science and Information Systems

2018 – 2022

October 6 University — Giza, Egypt

- Relevant coursework: Algorithms, Data Structures, Cloud Computing, Database Systems, Software Engineering.
- Graduation project (grade A): Learning Management System (Node.js, Express, MongoDB).